INFOSYS SPRINGBOARD

Documentation

Name:- Soham Badgujar

Project Title:- Interactive Analytics Dashboard for Global Income Distribution

Project Statement and Outcomes:-

The Interactive Analytics Dashboard for Global Income Distribution project aims to develop an interactive Power BI dashboard that visualizes income inequality data across various countries and regions. This project involves collecting, preprocessing, and structuring global income distribution and economic data to enable effective visualization. The dashboard will provide users with insights into income disparities over time, country-wise comparisons, and patterns of inequality globally.
The outcome will be a user-friendly dashboard, embedded within a Streamlit web application, offering an intuitive interface for economists, researchers, policymakers, and the general public. This platform will help users explore global income inequality, understand disparities across different regions, and analyze trends in wealth distribution. The project will conclude with comprehensive testing to ensure dashboard accuracy and usability, supported by detailed documentation for future reference and enhancements.

Data Sources and Acquisition

World Bank – World Development Indicators (WDI)

- The World Bank DataBank was used as a primary source for globally comparable economic and inequality indicators.

- Data Obtained
1. Gini Index (income inequality measure)
2. GDP per capita
3. Country and regional classifications
4. Year-wise economic indicators

- How the Data Was Collected
1. Visited the World Bank DataBank portal.
2. Selected the World Development Indicators (WDI) dataset.
3. Chose required indicators such as Gini Index and GDP per capita.
4. Filtered countries and years.
5. Exported the dataset in CSV format for further analysis.

- Reason for Use
- The World Bank provides officially validated and standardized global economic data widely used in research and policy analysis.

UNU-WIDER – World Income Inequality Database (WIID)

The WIID dataset was used to obtain detailed income distribution and inequality measurements collected from national statistical agencies.

- Data Obtained

1. Gini coefficients from multiple surveys
2. Income distribution indicators
3. Historical inequality records
4. Country-year observations

- How the Data Was Collected

1. Accessed the UNU-WIDER WIID database website.
2. Downloaded the latest WIID dataset release.
3. Selected relevant variables related to income inequality.
4. Imported the dataset into Power BI/Python for preprocessing.

- Reason for Use
- WIID provides one of the most comprehensive global inequality datasets, combining multiple national data sources across decades.

Kaggle Dataset Repository

Kaggle was used to obtain precompiled datasets that combine multiple global inequality indicators into structured formats.

- Data Obtained

1. Cleaned country-year datasets
2. Income distribution metrics
3. Supporting economic indicators

- How the Data Was Collected

1. Searched Kaggle using keywords such as global income inequality and Gini index dataset.
2. Selected datasets with sufficient country coverage and documentation.
3. Downloaded datasets in CSV format.
4. Validated and cleaned data before integration.

- Reason for Use
- Kaggle datasets helped accelerate development by providing already structured datasets suitable for visualization and exploratory analysis.

OECD Income Distribution Database (IDD)

The OECD IDD was used to supplement inequality data, particularly for developed economies.

- Data Obtained

1. Income distribution statistics
2. Disposable income inequality measures
3. Comparative indicators for OECD member countries

- How the Data Was Collected

1. Accessed the OECD Data portal.
2. Navigated to the Income Distribution Database (IDD) section.
3. Selected inequality indicators and country filters.
4. Exported datasets in CSV/Excel format.

- Reason for Use
- OECD data offers highly reliable and harmonized statistics, enabling accurate comparison among developed countries.

Data Cleaning and Master Sheet Creation

1. Cleaning & Standardization

- Gaps (1–2 years): Applied the Fill Down and Fill Up functions on the Gini and Income Share columns to propagate values from adjacent years.
- Gaps (3–10 years): For medium-to-long-term gaps, I used Group By operations to calculate the average value for that country across the decade, then merged this average back into the null fields using a conditional column.
- Gaps (>11 years): These were left as null to maintain data honesty, ensuring the dashboard accurately reflects periods where no global census or survey data exists.

Country Name Standardization and Cleaning:

Applied Transform > Trim and Transform > Clean to all country columns to eliminate trailing spaces and non-printable characters that cause merge errors.

- Standardization: Used Replace Values to unify naming conventions (e.g., changing "United States of America" and "USA" to a single "United States" format) ensuring 100% alignment across the three source sheets.

2. Merging & Master Sheet Creation

A. The Merge Process

Source Integration: Merged the GDP dataset (150 countries), Gini dataset (50 countries), and Income Share dataset (80 countries).

Method: Used "Merge Queries as New" with a Left Outer Join based on a composite key like of [Country Name] and [Year].

Target Alignment: The final Master Sheet was filtered to focus on ~50 countries where the intersection of indicators was strongest.

B. Final Master Sheet Refinement

Deduplication: Applied "Remove Duplicates" on the Country + Year pair to ensure each record is unique.

Dataset Scope: The Master Sheet successfully maintains data for over 20 countries across a 23-year span (2000–2023).

Column Optimization: Dropped redundant technical columns (like c3, c2, or internal IDs) to keep the dataset professional and lightweight for the Power BI engine.

3. Issues Faced & Resolutions

Issue: Inconsistent date formats across CSV files.
Resolution: Used Locale settings in the "Change Type" step to force all Year columns into a 4-digit Whole Number format.

Issue: Duplicate country entries in the WIID source.
Resolution: Filtered by quality_score in Power Query to retain only the most reliable record for the Master Sheet.